

PHYSICAL AI PROTOTYPE

TableTogether

Vision. Language. Coordinated action.



A table-setting assistant in simulation

**“Set the table.
Start with the cup.”**

Two SO-101 arms place a cup and utensil beside an existing plate.

The user can request either order.
The app shows live frames and placement checks.



Actual MuJoCo camera view

The learned perception-to-action policy

INPUTS

128 × 96 overhead RGB
Ordered instruction tokens
Twelve measured joint positions
Current action index

LEARNED OUTPUT

CNN and language encoder predict joint targets and durations over 18 keyframes.

MuJoCo actuators execute the targets through physical contacts.

Sequential complementary arm actions. No expert fallback during learned execution.

Training improved validation accuracy

Measure	Original	Revised
Successful demonstrations	43	75
Training keyframes	774	1,350
Validation joint MAE	0.002598 rad	0.001933 rad
Complete validation episodes	6/8	7/8

25.6% lower joint-target error on validation observations

Separate validation seeds. Fresh test scenes determine task reliability.

Fresh tests passed half the episodes

OpenVINO CPU condition	Successful episodes
Cup first	5/10
Utensil first	5/10

Ten fresh scenes, two instruction orders

Object positions vary independently by ± 8 mm.
Cup placement remains a frequent failure.

Better validation accuracy has not established reliable task completion.

Robustness beyond object placement

Scene condition	Success
Standard positions	6/10
Lighting and background	4/10
Mass and friction	6/10
Horizontal object size	7/10
Combined variations	6/10

50 additional episodes. Unfamiliar shapes and recovery remain untested.

Intel deployment: measured and expected

VERIFIED LOCALLY

Core Ultra 5 125U Series 1

MuJoCo simulation and OpenVINO CPU inference run locally.

All 112 model graph operations pass the CPU support query.

No full-policy NPU or GPU execution claim

TARGET HARDWARE

Core Ultra Series 2 / 3

Intel's CPU support matrix supports compatibility in principle.

Actual target execution and performance remain unverified.

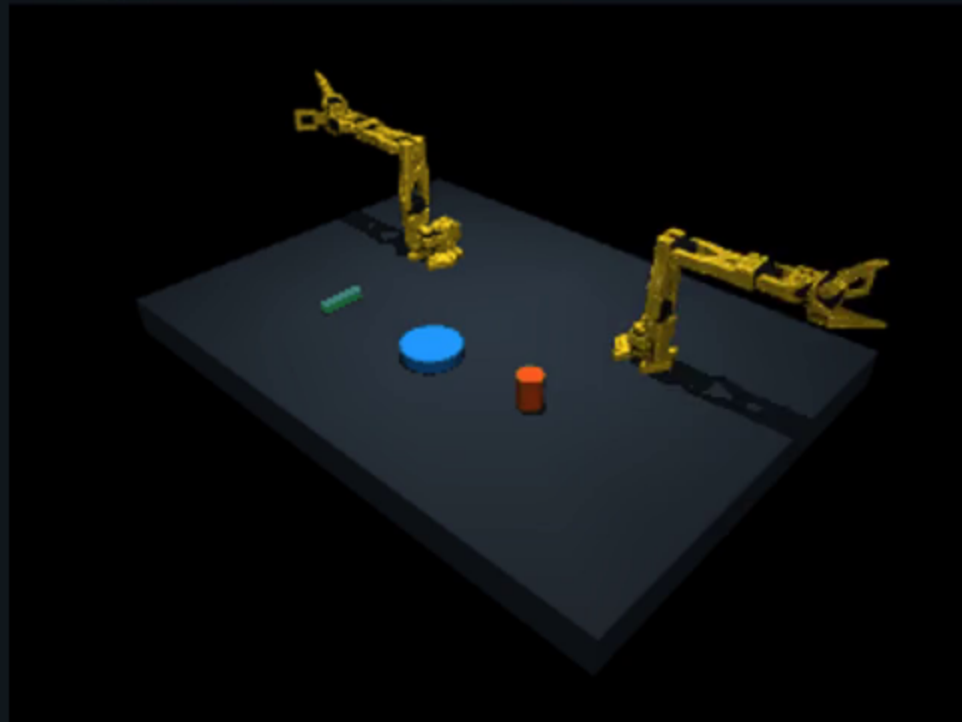
The complete ten-seed demonstration

TableTogether | Seed 600 | openvino CPU | ~2x

Command: Set the table. Start with the cup.

XY +/-8 mm | Frozen model f2f60dde4d38

Actual contact simulation; no expert fallback



2 minutes 37 seconds

All ten consecutive seeds
Every success and failure
Visible command and model identity

[Watch the recorded demonstration](#)

[Open the source repository](#)

Next steps toward a qualifying submission

Improve contact reliability

Collect failure-recovery demonstrations and evaluate closed-loop correction.

Validate on Series 2 or Series 3

Run simulation, record hardware identity, and benchmark the complete policy.

github.com/SoniaLong/table-together

Source, training data, model weights and evaluation evidence are public.